Substantial and Partial Derivatives

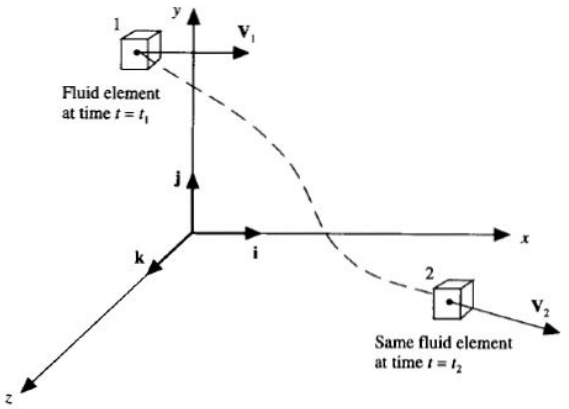
Consider an infinitesimally small fluid element moving with the flow through cartesian space

The unit vectors along the x, y, z are i, j, k resp.

The ~~vector~~ velocity vector field in the cartesian space.

$$V = u\hat{i} + v\hat{j} + w\hat{k}$$

$$u = u(x, y, z, t)$$
$$v = v(x, y, z, t)$$
$$w = w(x, y, z, t)$$



Considering unsteady flow,

Scalar density field

$$\rho = \rho(x, y, z, t)$$

At Time t_1

$$\rho_1 = \rho(x_1, y_1, z_1, t_1)$$

At later Time t_2,

$$\rho_2 = \rho(x_2, y_2, z_2, t_2)$$

Expanding this function in Taylor Series about point 1.

$$\rho_2 = \rho_1 + \left(\frac{\partial \rho}{\partial x}\right)_1 (x_2 - x_1) + \left(\frac{\partial \rho}{\partial y}\right)_1 (y_2 - y_1) + \left(\frac{\partial \rho}{\partial z}\right)_1 (z_2 - z_1) + \left(\frac{\partial \rho}{\partial t}\right)_1 (t_2 - t_1) + \text{Higher Order}$$

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Dividing $(t_2 - t_1)$ and Ignoring Higher Orders

$$\frac{\rho_2 - \rho_1}{t_2 - t_1} = \left(\frac{\partial \rho}{\partial x}\right)_1 \frac{x_2 - x}{t_2 - t_1} + \left(\frac{\partial \rho}{\partial y}\right)_1 \frac{y_2 - y_1}{t_2 - t_1} + \left(\frac{\partial \rho}{\partial z}\right) \frac{z_2 - z_1}{t_2 - t_1} + \left(\frac{\partial \rho}{\partial t}\right)_1$$

This is physically the average time rate of change in Density of the fluid element as it moves from 1 to 2.

$$\lim_{t_2 \to t_1} \frac{\rho_2 - \rho_1}{t_2 - t_1} = \frac{D\rho}{Dt}$$

$\frac{D\rho}{Dt}$ → is the instantaneous time rate of change of density.
This is called as Substantial Derivative.

$\frac{D}{Dt}$ → Time rate of change of density of fluid element as it moves through space.

$\frac{\partial \rho}{\partial t}$ → Partial Derivative.

$\frac{\partial}{\partial t}$ → Time rate of change of density at a fixed point

$$\frac{D\rho}{Dt} = u\frac{\partial \rho}{\partial x} + v\frac{\partial \rho}{\partial y} + w\frac{\partial \rho}{\partial z} + \frac{\partial \rho}{\partial t}$$

Substantial Derivative in cartesian co-ordinate,

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y} + w\frac{\partial}{\partial z}$$

∇ = Vector Operator $= i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + (V \cdot \nabla) \quad \rightarrow \text{In Vector Form}$$

Date / / Page Clevermate

Acceleration

Acceleration can be defined as the rate of change of velocity with respect to time.

Velocity, $\vec{V} = u(x,y,z,t)\,\hat{i} + v(x,y,z,t)\,\hat{j} + w(x,y,z,t)\,\hat{k}$

$u, v, w \rightarrow$ components of velocity in x, y, z direction

$$a = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$a_x = \lim_{dt \to 0} \frac{du}{dt}$$

$$a_y = \lim_{dt \to 0} \frac{dv}{dt}$$

$$a_z = \lim_{dt \to 0} \frac{dw}{dt}$$

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} + \frac{\partial u}{\partial z}\frac{dz}{dt} + \frac{\partial u}{\partial t}\frac{dt}{dt}$$

$$\lim_{dt \to 0} \frac{du}{dt} = \frac{\partial u}{\partial x}u + \frac{\partial u}{\partial y}v + \frac{\partial u}{\partial z}\cdot w + \frac{\partial u}{\partial t}$$

$$a_x = u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} + \frac{\partial u}{\partial t}$$

Similarly,

$$a_y = u\cdot\frac{\partial v}{\partial x} + v\cdot\frac{\partial v}{\partial y} + w\cdot\frac{\partial v}{\partial z} + \frac{\partial v}{\partial t}$$

$$a_z = u\cdot\frac{\partial w}{\partial x} + v\cdot\frac{\partial w}{\partial y} + w\cdot\frac{\partial w}{\partial z} + \frac{\partial w}{\partial t}$$

$$a = \frac{D\vec{V}}{Dt} = \frac{\partial \vec{V}}{\partial t} + \vec{V}\cdot(\vec{V}\cdot\vec{\nabla})$$

$\frac{\partial V}{\partial t} \rightarrow$ Local Acceleration

$\vec{V}\cdot(\vec{V}\cdot\vec{\nabla}) \rightarrow$ Convective Acceleration.

$$\vec{a} = \frac{d\vec{V}}{dt}$$

$$\vec{V} = u\hat{i} + v\hat{j} + w\hat{k}$$

$$\frac{d\vec{V}}{dt} = \underbrace{u\frac{\partial \vec{V}}{\partial x} + v\frac{\partial \vec{V}}{\partial y} + w\frac{\partial \vec{V}}{\partial z}}_{\text{Convective Acceleration}} + \underbrace{\frac{\partial \vec{V}}{\partial t}}_{\text{Local Acceleration}}$$

$$\left.\begin{aligned} &\frac{\partial u}{\partial t} \\ &\frac{\partial v}{\partial t} \\ &\frac{\partial w}{\partial t} \end{aligned}\right\} \text{Local Acceleration}$$

$$\left.\begin{aligned} &u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} \\ &u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} \\ &u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} \end{aligned}\right\} \text{Convective Acceleration}$$

$$\vec{a}_{total} = \vec{a}_{conv.} + \vec{a}_{local.}$$